

StyleMate Ecosystem

Target Environment: Private App Playground / Vercel

Primary Accent: Royal Blue (#1a5fd4)

Version: 1.0

1. Project Overview & Context

StyleMate is a private, backend-hosted utility dashboard built seamlessly into an existing personal portfolio platform ecosystem. It serves as an automated personal wardrobe tracker, environmental readiness monitor, and lifestyle inventory assistant accessible via internal Google OAuth. The primary goal is to minimize daily cognitive friction associated with morning coordination and weather preparation while cataloging personal fashion trends natively over time.

2. Core Architecture & Features

2.1 Manual Closet Scanner & Quick-Log

A highly streamlined, mobile-optimized submission portal configured for the rapid initial configuration and bulk parsing of a personal clothing inventory. Supports scanning attributes, categories, and direct photo logs to preserve high-fidelity visual representations of individual items.

2.2 Gmail API Automated Purchase Detector

Asynchronous background integration engine leveraging existing Google authentication privileges to process incoming purchase confirmations from e-commerce merchants. Detects digital orders in real time to automatically prompt staging queues for newly acquired apparel.

2.3 Hyperlocal Weather & Environmental Readiness Banner

A real-time telemetry strip communicating immediate atmospheric conditions, local temperature bands, and absolute rain/precipitation threshold notifications before leaving the workspace environment.

3. Detailed Technical Specifications

3.1 Database Relational Schema

The layout data model requires precise categorical classifications, operational ranges, and strict lifecycle states to handle predictive lookups in future updates:

```

CREATE TYPE apparel_category AS ENUM ('top', 'bottom', 'outerwear', 'footwear', 'accessory');
CREATE TYPE laundry_status AS ENUM ('clean', 'dirty', 'laundry_cycle');

CREATE TABLE wardrobe_items (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(255) NOT NULL,
  category apparel_category NOT NULL,
  sub_type VARCHAR(100), -- e.g., 'short_sleeve', 'shorts', 'hoodie'
  color VARCHAR(50) NOT NULL, -- e.g., 'royal_blue', 'cream', 'black'
  min_temp_threshold INT DEFAULT 0, -- Min temp required to wear comfortably
  max_temp_threshold INT DEFAULT 110,
  archetype VARCHAR(100) DEFAULT 'casual_weekend', -- 'gym_grind', 'client_meeting', 'old_n
  status laundry_status DEFAULT 'clean',
  is_incoming BOOLEAN DEFAULT FALSE, -- Set true if parsed automatically from Gmail
  image_url TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

3.2 Weather Analysis & Threshold Rules

- **Integration Endpoint:** OpenWeatherMap API (One Call API 3.0 via Free Tier allocations).
- **The Umbrella Threshold:** If hourly precipitation_probability (pop) registers above **30%** during active daylight intervals, toggle a persistent visual alert bar across the primary panel display.
- **Spring / Summer Thermal Profiling:**
 - When the localized ambient temperature registers > **60°F**, the UI matrix prioritizes short sleeves, shorts, and active lightweight selections.
 - When temperatures scale below **60°F**, the dashboard shifts to expose verified clean outerwear layering solutions.

3.3 Gmail API Pipeline Mechanics

To capture physical additions without polling overhead, a serverless background worker triggers against specific parameters:

- **Pipeline Filter:** Targets mail headers containing from:amazon.com "Your Purchase" or from:depop.com "Your Order".
- **The Staging Queue:** Parsed names are safely committed directly to the database with the structural flag `is_incoming = true`.
- **Interactivity:** Upon subsequent app initialization, a glassmorphic confirmation modal surfaces: *"We detected new items on their way! Click to officially approve them into your layout matrix."*



System Layout Guidance for Cursor Integration

Instruct Cursor to build the interface using standard web container boundaries wrapped completely inside background-blur utilities (`backdrop-blur-md bg-white/10 border border-white/20`). Maintain visual hierarchy using a top-anchored environmental emergency strip, a multi-column closet organizer grouping items into distinct aesthetic tracks (Gym/Grind, Client Meeting, Old Money), and a dedicated drag-and-drop lookbook editor panel.

4. Edge Cases & Safety Fallbacks

- **Laundry Depletion Protocol:** If total verified clean clothing entries inside core categories (outerwear or bottoms) diminish below a threshold value of 2, append a soft visual indicator badge warning: ⚠️ **Wardrobe Friction: 80% of outerwear is currently marked as dirty.**
- **API Failovers:** If the external weather lookup encounters an unhandled timeout or network fault, gracefully fall back to historical seasonal averages based on regional parameters without throwing structural frontend errors.

5. Future Architectural Expansion Roadmap

Engine Paradigm	Algorithmic Approach	Future Stack Prerequisites
Heuristic Rule Matrix	Hardcoded mathematical filters validating color harmony pairs and strict threshold bounds.	TypeScript Vector Arrays / Cosine Similarity Filters
Autonomous LLM Agent	Serializing the current clean inventory table state as a contextual JSON payload fed directly into a lightweight completion model.	Serverless OpenAI / Gemini API Completion Tasks